



# Potenciál elektrického dipólu

el. dipól

dipólový moment:  $\vec{p} = Q\vec{d}$

elementární dipól:  $\vec{p} = \hbar e \vec{\sigma}$ ,  $|\vec{d}| \rightarrow 0$  Bodový, elementární dipól

EL. POLE DIPÓLU

$\vec{r}^- + \vec{d} = \vec{r}^+$

$\varphi(\vec{r}) = \frac{1}{4\pi\epsilon_0} \left( \frac{Q}{|\vec{r} - \vec{r}^-|} - \frac{Q}{|\vec{r} - \vec{r}^+|} \right)$

$\varphi = \frac{1}{4\pi\epsilon_0} Q \left( \frac{1}{R^+} - \frac{1}{R^-} \right)$

$\varphi = \frac{Q}{4\pi\epsilon_0} \frac{R^- - R^+}{R^+ R^-}$

EL. DIPÓL:

$R^+ R^- = (R - \frac{d}{2} \cos \vartheta) (R + \frac{d}{2} \cos \vartheta)$

$R^2 - \frac{d^2}{4} \cos^2 \vartheta$

$R^- = R^+ + d \cos \vartheta$

$\varphi = \frac{1}{4\pi\epsilon_0} \frac{Q d \cos \vartheta}{R^2} = \frac{1}{4\pi\epsilon_0} \frac{\vec{p} \cdot \vec{r}}{R^3}$

$\cos \vartheta = \frac{\vec{p} \cdot \vec{r}}{p r}$

BODOVÝ DIPÓL:

$\varphi = \frac{1}{4\pi\epsilon_0} \frac{\vec{p} \cdot \vec{r}}{r^3}$

# Elektrické pole elektrického bodového dipólu

Bodový dipól:  $\varphi = \frac{1}{4\pi\epsilon_0} \frac{\vec{p} \cdot \vec{r}}{r^3}$   
(v rovině TKH)

~ pole  $\vec{E}(\vec{r})$   $\vec{E} = -\nabla\varphi$

$$\nabla\varphi = \frac{1}{4\pi\epsilon_0} \nabla \left( \frac{\vec{p} \cdot \vec{r}}{r^3} \right)$$

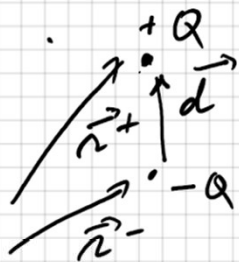
$$\underline{x}: (\nabla\varphi)_x = \frac{\partial}{\partial x} \left( \frac{p_x x + p_y y + p_z z}{(x^2 + y^2 + z^2)^{3/2}} \right) = -\frac{3}{2} (x^2 + y^2 + z^2)^{-5/2} 2x (\vec{p} \cdot \vec{r})$$

$$+ (x^2 + y^2 + z^2)^{-3/2} p_x = -\frac{3x(\vec{p} \cdot \vec{r})}{r^5} + \frac{p_x}{r^3}$$

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \left[ \frac{3\vec{r}(\vec{p} \cdot \vec{r})}{r^5} - \frac{\vec{p}}{r^3} \right]$$

# Energie elektrického dipólu ve vnějším elektrickém poli

ENERGIE DIPÓLU V EL. POLI



$$W = Q\varphi(\vec{r}^+) + (-Q)\varphi(\vec{r}^-)$$

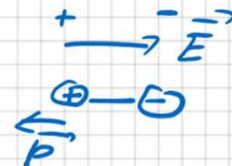
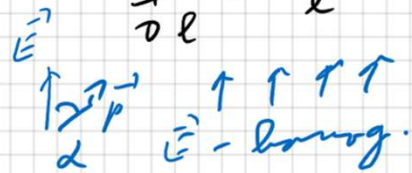
$$W = Q [\underbrace{\varphi(\vec{r}^+) - \varphi(\vec{r}^-)}_{\vec{d} \cdot \nabla \varphi}] = \underbrace{Q\vec{d}}_{\vec{p}} \cdot \nabla \varphi$$

derivace se směr:

$$\frac{\partial \varphi}{\partial l} = \frac{\vec{l}}{l} \cdot \nabla \varphi$$

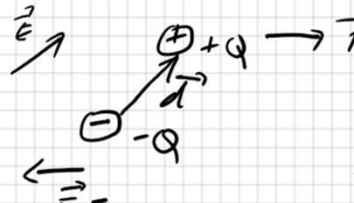
$$| W = \vec{p} \cdot \nabla \varphi = - \vec{p} \cdot \vec{E} |$$

$$W = -pE \cos \alpha$$



Síla působící na elektrický dipól ve vnějším elektrickém poli

EL. DIPÓL V EL. POLI  $\vec{E}$  ?  $\vec{F}$   
SÍLA



$$\vec{F} = \vec{F}^+ + \vec{F}^- = Q\vec{E}(\vec{r}^+) + (-Q)\vec{E}(\vec{r}^-)$$

$$\vec{F} = Q[\vec{E}(\vec{r}^+) - \vec{E}(\vec{r}^-)]$$

$$F_x = Q[E_x(\vec{r}^+) - E_x(\vec{r}^-)] = Q\vec{d} \cdot \nabla E_x = (\vec{p} \cdot \nabla) E_x$$

$\vec{d} \cdot \nabla E_x$        $\vec{p}$        $p_x \frac{\partial}{\partial x} + p_y \frac{\partial}{\partial y} + p_z \frac{\partial}{\partial z}$

$$\vec{r}^+ - \vec{r}^- = \vec{d} \quad | \quad \vec{F} = (\vec{p} \cdot \nabla) \vec{E} |$$

Pozn.  $F_x = (\vec{p} \cdot \nabla) E_x = p_x \frac{\partial}{\partial x} E_x + p_y \frac{\partial}{\partial y} E_x + p_z \frac{\partial}{\partial z} E_x =$

! Než  $\vec{E} = 0$ !  $\frac{\partial E_x}{\partial y} = \frac{\partial E_y}{\partial x}, \frac{\partial E_x}{\partial z} = \frac{\partial E_z}{\partial x} \quad \left| \quad p_x \frac{\partial}{\partial x} E_x + p_y \frac{\partial E_x}{\partial x} + p_z \frac{\partial E_x}{\partial x} \right|$

$$= \frac{\partial}{\partial x} (p_x E_x + p_y E_y + p_z E_z)$$

$$= \frac{\partial}{\partial x} (\vec{p} \cdot \vec{E})$$

$$| \vec{F} = \nabla (\vec{p} \cdot \vec{E})$$

$$\vec{F} = -\nabla W = -\nabla (-\vec{p} \cdot \vec{E})$$

# Dipól v homogenním poli – moment dvojice sil

